

Monopoly and Market Power

Lecture 6

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Sources of Monopoly Power

Why are some markets served by a single firm?

Technical / Cost Barriers

- **Economies of scale:** average cost falls over the relevant range of demand → natural monopoly
- **Control of key inputs:** e.g., De Beers and diamond mines (historically)
- **Network effects:** value rises with users (platforms, operating systems)

Legal / Strategic Barriers

- **Patents & IP:** pharmaceutical exclusivity periods
- **Government franchises:** postal services, utilities
- **Licensing & regulation:** spectrum allocation, taxi medallions

Strategic Behavior

- **Predatory pricing:** price below cost to deter entry
- **Exclusive contracts:** lock up distribution channels
- **Excess capacity:** credible threat of aggressive response

Applied Note

Modern antitrust debates focus on whether tech platforms (Google, Amazon, Meta) constitute monopolies through network effects and data advantages.

⋮

Examples

Can you think about some companies that have monopoly power? What are the sources of their market power?

Application: *U.S. v. Google* (2024–25)

The Case

In August 2024, Judge Amit Mehta ruled that Google holds **monopoly power** in general search (~90% desktop, ~95% mobile) and used **exclusive distribution contracts** to maintain it — violating §2 of the Sherman Act.

The Monopoly Maintenance Mechanism

Google paid Apple an estimated **\$26B/year** to be the default search engine on Safari/iOS. The court found these contracts:

- Foreclosed rivals (Bing, DuckDuckGo) from the scale needed to improve quality

Remedies (Sept. 2025)

Court **rejected** structural breakup but imposed **behavioral remedies**:

- Banned exclusive default search contracts
- Required data-sharing with competitors
- Prohibited bundling of Chrome and AI products

Discussion Questions

- Is Google's dominance due to a superior product or exclusionary contracts? How would you test this?
- Will behavioral remedies suffice, or does the data-feedback loop recreate monopoly?

Natural Monopoly

Definition

A **natural monopoly** exists when a single firm can supply the entire market at lower cost than two or more firms — the cost function is **subadditive**:

$$C(Q) < C(Q_1) + C(Q_2) \quad \text{for all } Q_1 + Q_2 = Q$$

Sufficient (not necessary) condition: economies of scale over the entire relevant range, i.e., $AC(Q)$ is declining at market demand.

Classic Examples

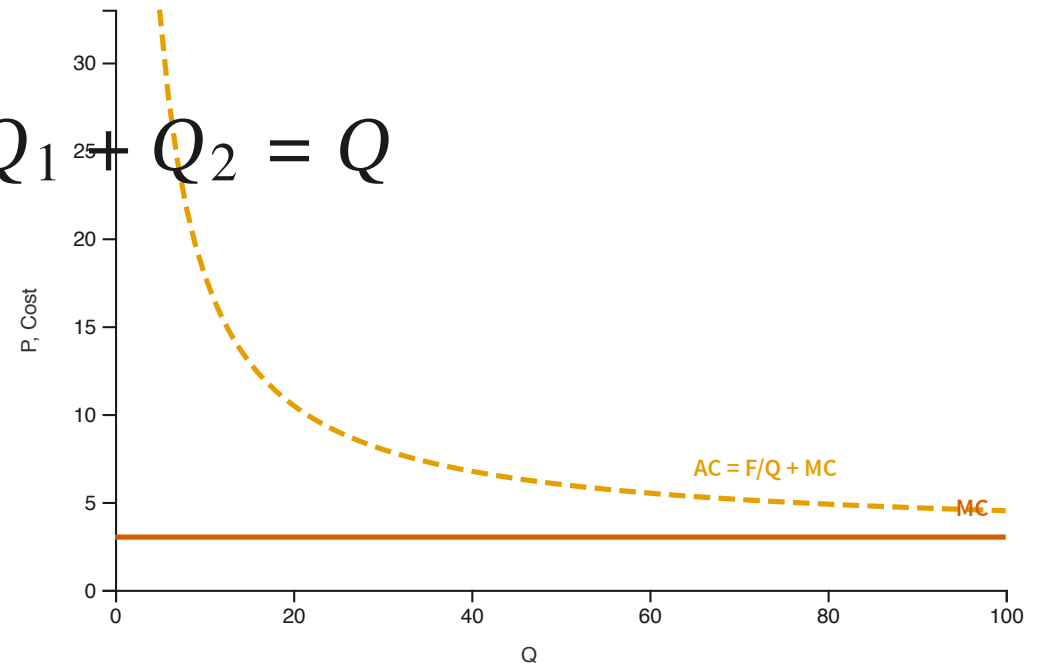
- Electricity transmission & distribution
- Water and sewer systems

Fixed Cost (F)

150

MC

3



Monopoly Profit Maximization

The monopolist's problem: choosing output (or price) to maximize profit.

The Monopolist's Problem

The Monopolist's Problem

The monopolist faces the *entire* market demand $P(Q)$. Unlike a competitive firm, selling more requires lowering price.

$$\max_Q \pi(Q) = P(Q) \cdot Q - C(Q)$$

First-Order Condition

$$\frac{d\pi}{dQ} = P(Q) + Q \cdot P'(Q) - C'(Q) \stackrel{!}{=} 0$$

$$\boxed{MR(Q) = MC(Q)}$$

Marginal Revenue Decomposition

$$MR = P + Q \cdot \frac{dP}{dQ}$$

Result

Since $dP/dQ < 0$, we always have $MR < P$. The monopolist produces where $MR = MC$, then charges $P^* = P(Q^*) > MC$.

The Inverse Elasticity Rule

$$MR = P \left(1 + \frac{1}{\varepsilon_D} \right) = P \left(1 - \frac{1}{|\varepsilon_D|} \right)$$

Setting $MR = MC$:

$$\boxed{\frac{P - MC}{P} = \frac{1}{|\varepsilon_D|}}$$

The **Lerner Index** (markup over price)

Monopoly Equilibrium — Interactive

Demand intercept (a)

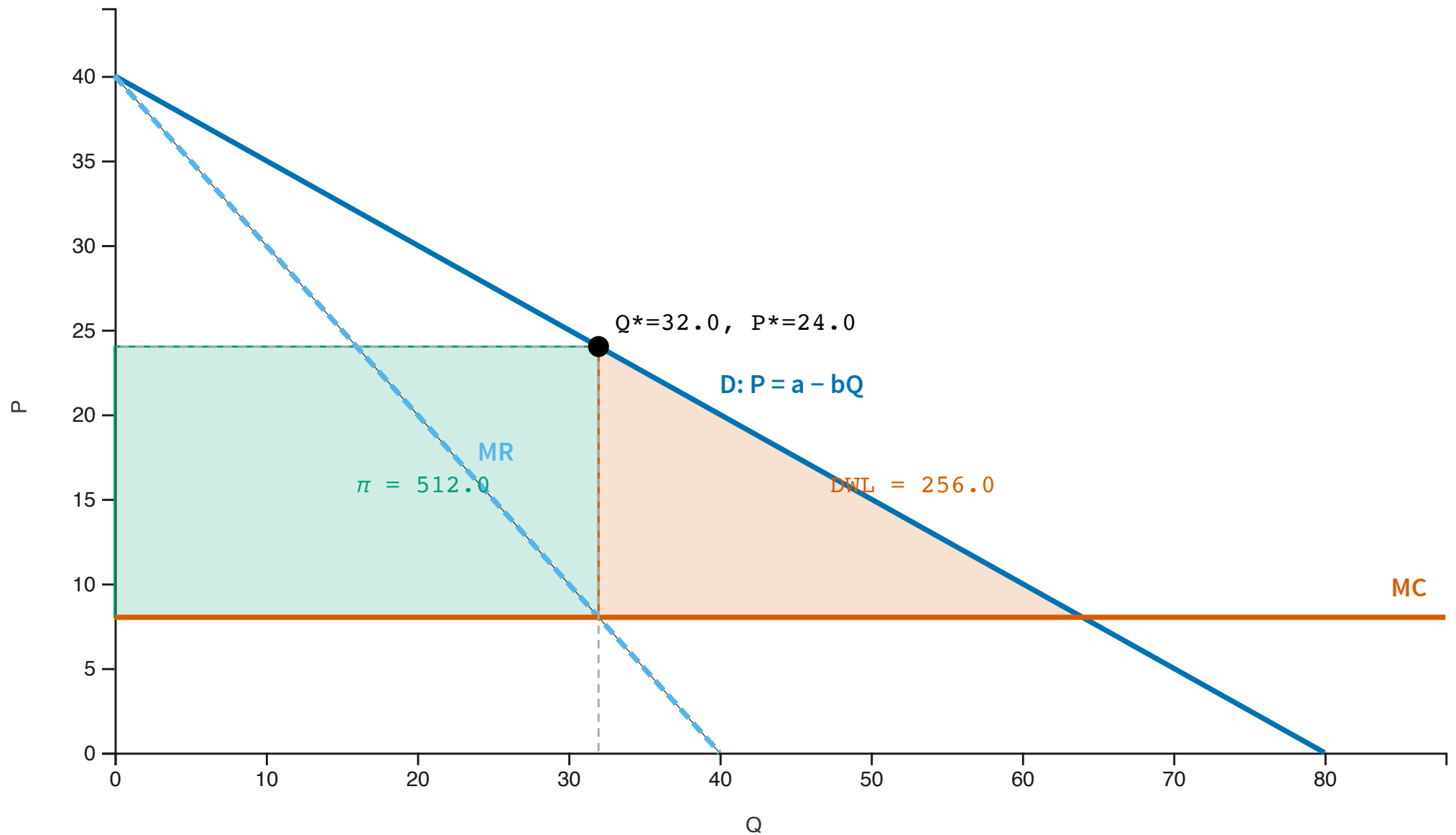


Demand slope (b)



MC (c)





Linear demand: $P = a - bQ$, constant $MC = c$. ■ monopoly profit, ■ deadweight loss.

Why Monopolists Have No Supply Curve

Key Insight

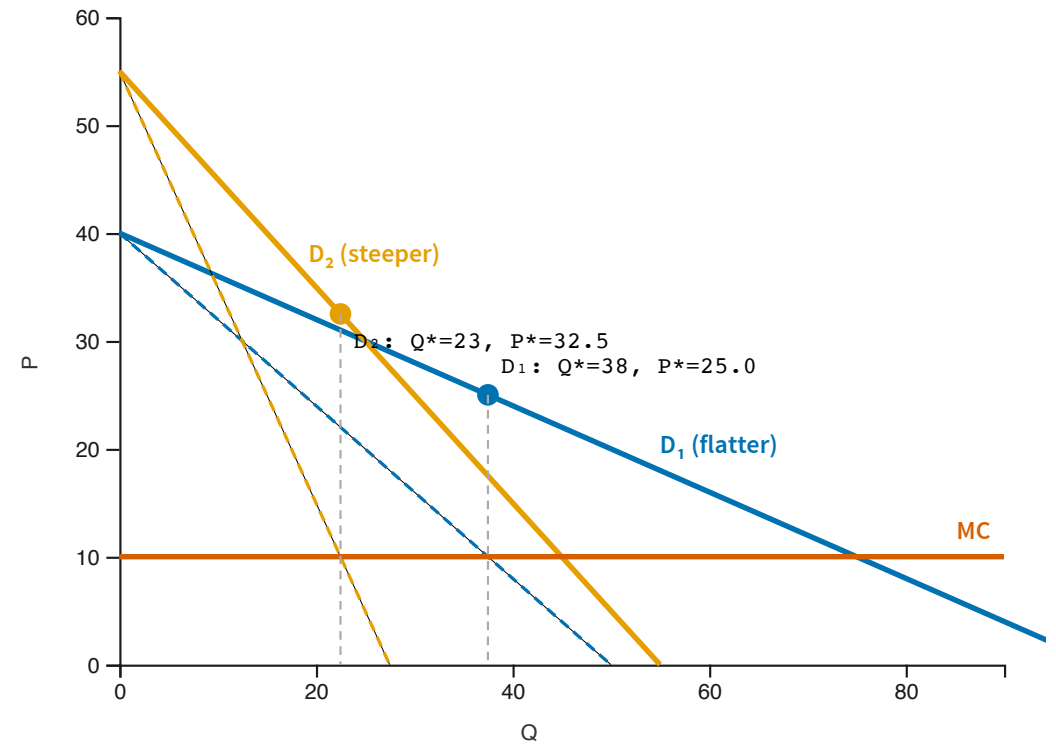
A supply curve maps each price to a unique quantity. For a monopolist, the **optimal quantity depends on the shape of the demand curve**, not just the price level.

Two different demand curves can intersect the same MC at the same price but yield different monopoly quantities — because MR depends on the slope/elasticity of demand, not just its level.

Formally

Q^* such that $MR(Q^*) = MC(Q^*)$

Since MR depends on the entire demand function (not just P) there is



Two demand curves yield the same price but different quantities — no unique supply relationship.

Welfare and Deadweight Loss

Quantifying the social cost of market power.

Deadweight Loss of Monopoly

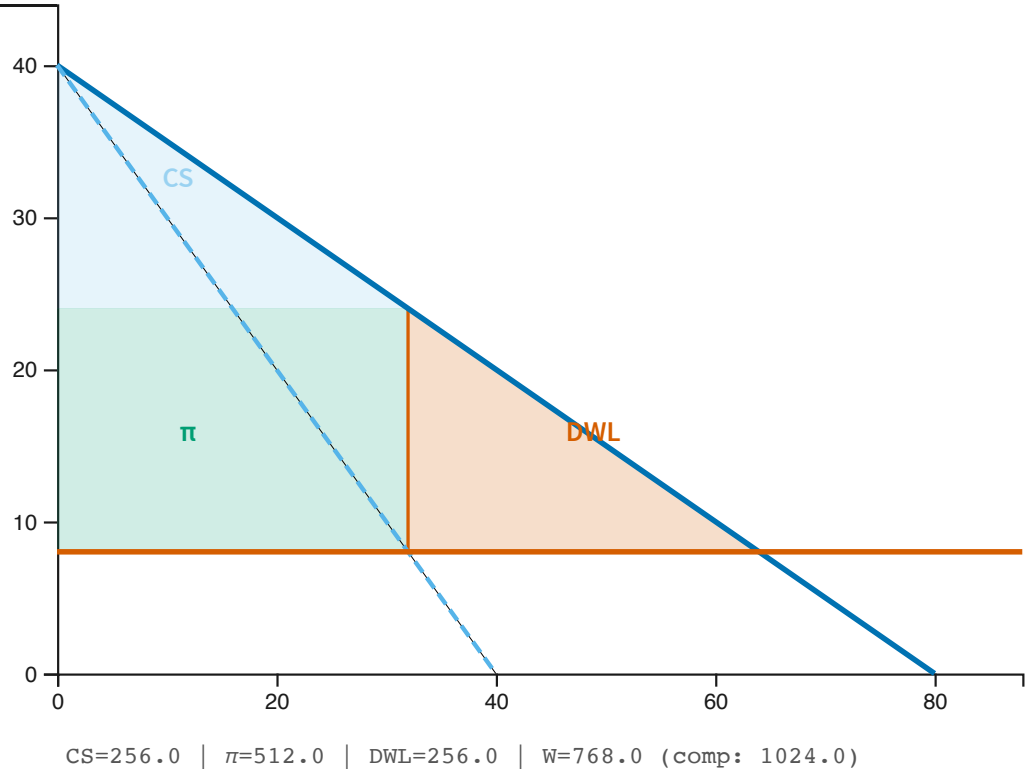
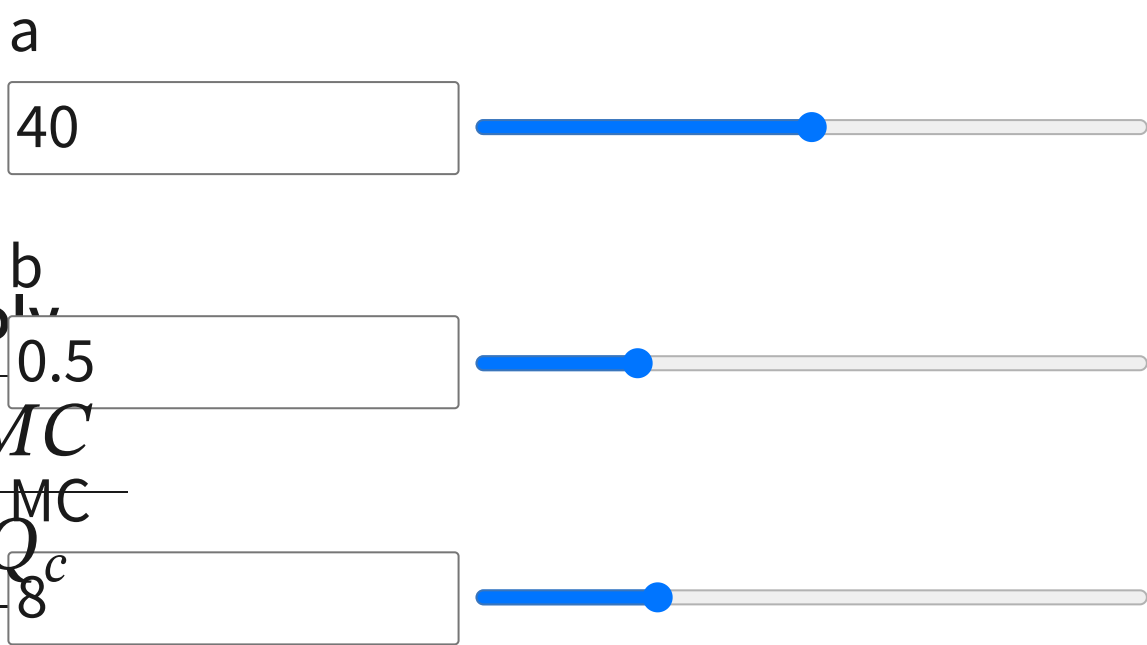
Compare monopoly to the competitive benchmark ($P = MC$):

	Competition	Monopoly
Price	$P_c = MC$	$P_m > MC$
Quantity	Q_c	$Q_m < Q_c$
CS	Large	Smaller
PS	Zero (LR)	Positive (profit)

Deadweight Loss

DWL = value of units between Q_m and Q_c that consumers value above MC but that are not produced.

With Linear Demand



Rent-Seeking and the Full Social Cost

Rent-Seeking (Tullock, 1967; Posner, 1975)

Firms spend real resources to acquire or maintain monopoly power. These expenditures are a *pure social waste* — they don't create value, just redistribute it.

Examples

- Lobbying for protective regulation
- Litigation to extend patents
- Campaign contributions for favorable policy
- Excess capacity as entry deterrence

Posner's Argument

In the limit, firms compete away the *entire profit rectangle* in rent-seeking.

Empirical Estimates of DWL

Harberger (1954) estimated US monopoly DWL at about **0.1% of GDP** — surprisingly small.

Why So Small?

- Used average industry elasticities → understates true DWL
- Assumed constant MC
- Did not account for rent-seeking
- Later estimates (Cowling & Mueller, 1978) found 4–13% of corporate value added

X-Inefficiency (Leibenstein, 1966)



Application: The Rise of Markups

De Loecker, Eeckhout & Unger (QJE 2020)

The Paper

Using Compustat firm-level data (US, 1955–2016), they estimate markups via the **production function approach** (Hall 1988): $\text{markup} = \text{output elasticity of variable input} \div \text{expenditure share}$.

Key Findings

- Aggregate markups rose from **~1.21 in 1980 to ~1.61 in 2016**
- Average profit rate rose from **1% to 8%**
- Increase driven by the **upper tail** — median unchanged
- Rise occurs **within industries**, not between them
- Reallocation of market share from low- to high-markup firms

The Ongoing Debate

Market Power View

Rising markups reflect **increased market power** — weaker antitrust, network effects, “winner-take-most” in tech.

Implies: welfare loss, stronger competition policy needed.

Technology/Efficiency View

Rising markups reflect **rising fixed costs** (R&D, software, intangibles). High markups compensate for overhead.

Implies: Schumpeterian rents, may be welfare-enhancing.

Fed Research (2024)

Comparative Statics

How do changes in demand or costs shift monopoly outcomes?

Comparative Statics of Monopoly

Shift in Demand

With linear demand $P = a - bQ$ and constant $MC = c$:

$$Q^* = \frac{a - c}{2b}, \quad P^* = \frac{a + c}{2}, \quad \pi^* = \frac{(a - c)^2}{4b}$$

An increase in a (demand shifts out):

- $\partial Q^* / \partial a = 1/(2b) > 0$
- $\partial P^* / \partial a = 1/2$ — passes through exactly half
- $\partial \pi^* / \partial a = (a - c)/(2b) > 0$

Shift in Cost

An increase in c (MC rises):

- $\partial Q^* / \partial c = -1/(2b) < 0$
- $\partial P^* / \partial c = 1/2$ — absorbs half the cost increase
- $\partial \pi^* / \partial c = -(a - c)/(2b) < 0$

Key Result: Pass-Through

With linear demand and constant MC, the monopolist's **pass-through rate is 50%**. Only half of a unit tax is borne by consumers.

Caution

These clean results depend on linearity. With iso-elastic demand, pass-through can exceed 100%. With convex demand, pass-through > 50%

Effect of a Unit Tax — Interactive

a

40

A horizontal slider bar with a blue dot at approximately 75% of the range.

b

0.5

A horizontal slider bar with a blue dot at approximately 25% of the range.

MC

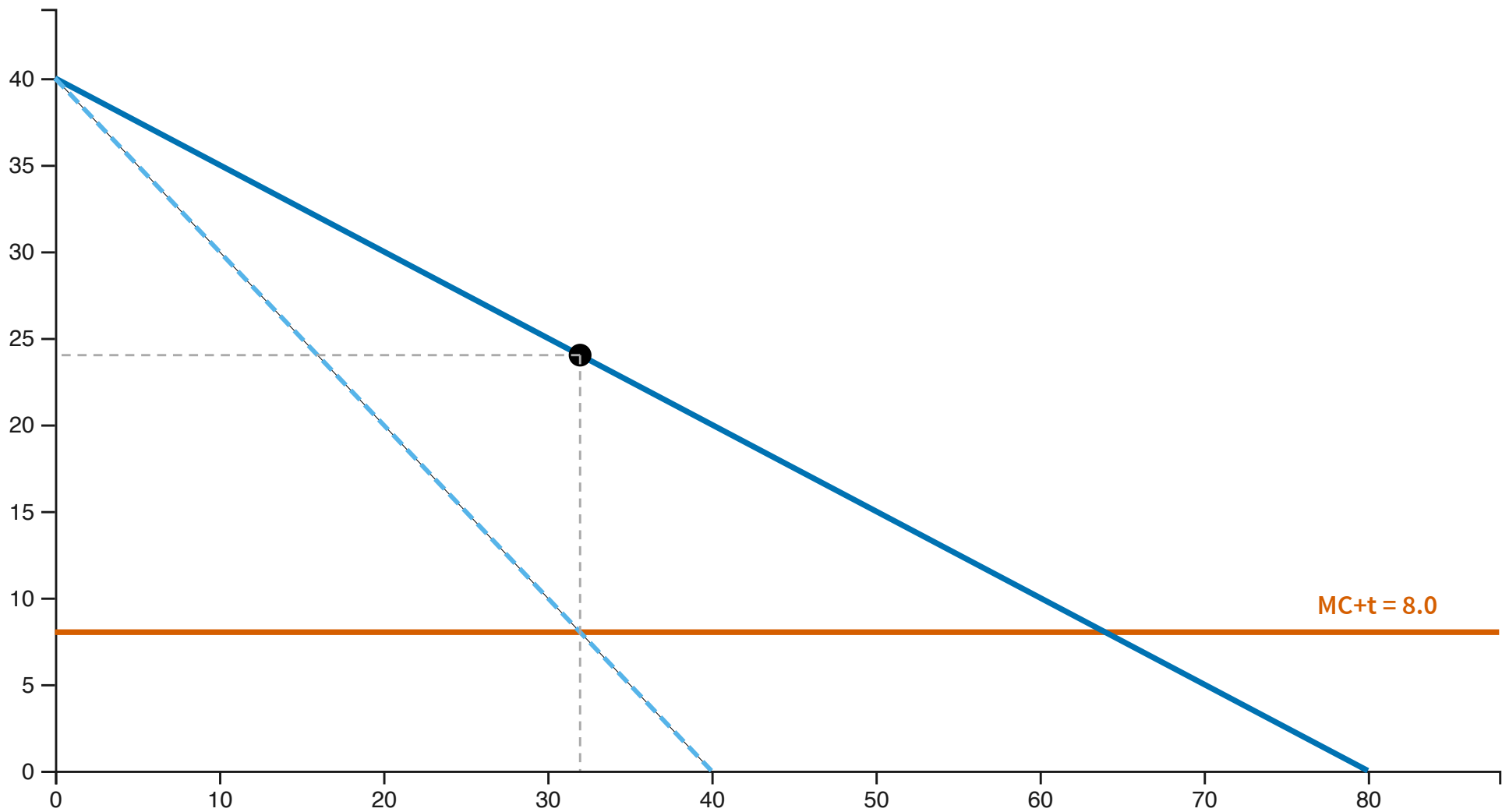
8

A horizontal slider bar with a blue dot at approximately 50% of the range.

Tax (t)

0

A horizontal slider bar with a blue dot at the far left (0% of the range).



No tax applied. Adjust slider.

Price Discrimination

Charging different prices to different consumers (or for different units).

Three Degrees of Price Discrimination

First Degree (Perfect)

Charge each consumer their max WTP for each unit.

Result: Extracts all CS.
Output = Q_c . No DWL — but all surplus goes to firm.

Requires perfect info.
Rare in practice.

Second Degree (Screening)

Offer a **menu** and let consumers self-select.
Price varies by *quantity* or *version*.

Mechanisms: Block pricing, bundling, versioning (airline classes), two-part tariffs.
Uses incentive compatibility — no need to identify types.

Third Degree (Segmentation)

Different prices to **identifiable groups** with different elasticities.

Rule: Higher markup in less elastic market.

$P_1/P_2 = (1 - 1/|\varepsilon_2|)/(1 - 1/|\varepsilon_1|)$
Student discounts, geographic pricing, peak/off-peak.

Requirements

1. Market power. (2) Ability to sort/identify consumers. (3) No arbitrage — consumers can't resell from low- to high-price segment.

Third-Degree Price Discrimination — Interactive

a_1 (Market 1)



b_1



a_2 (Market 2)



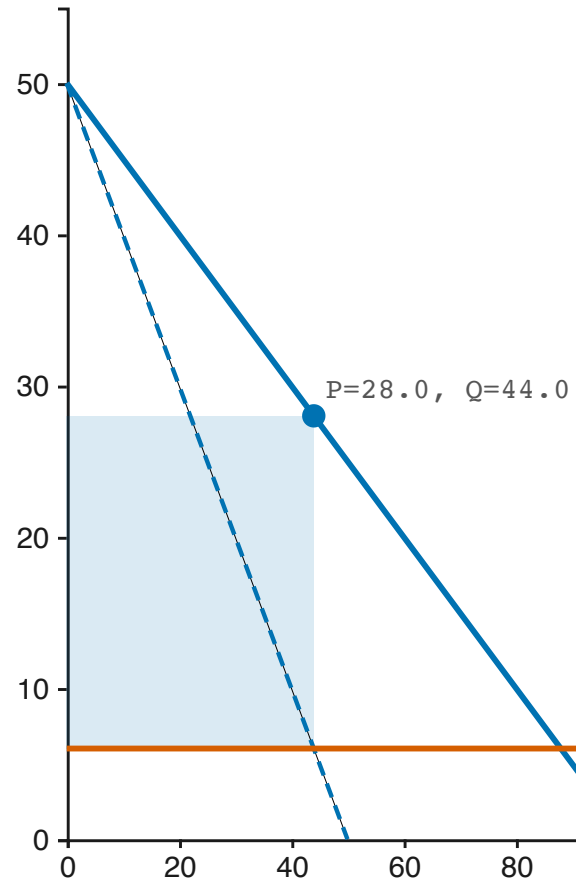
b_2



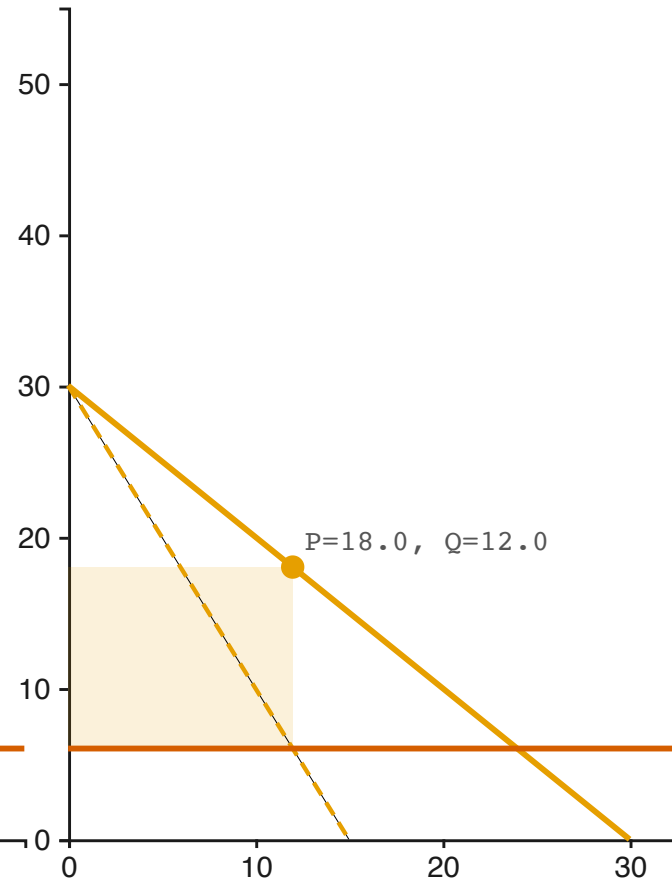
MC



Market 1 (less elastic)



Market 2 (more elastic)



Summary

Market 1: $P_1=28.0$, $Q_1=44.0$

$|\varepsilon_1| = 1.27$

$\pi_1 = 968.0$

Market 2: $P_2=18.0$, $Q_2=12.0$

$|\varepsilon_2| = 1.50$

$\pi_2 = 144.0$

Total $\pi = 1112.0$

Total Q = 56.0

$Lerner_1 = 0.786$

$Lerner_2 = 0.667$

Two-Part Tariffs

Mechanism

Charge a **fixed fee** (entry/access charge) T plus a **per-unit price** p .

Total payment: $T + p \cdot q$

With Identical Consumers

Set $p = MC$ (efficient usage) and $T = CS(MC)$ to extract all surplus.

$$\pi = N \cdot CS(MC) + \underbrace{(MC - MC)}_{=0} \cdot Q = N \cdot T$$

This replicates first-degree price discrimination!

With Heterogeneous Consumers

Real-World Examples

Setting	Fixed Fee	Per-Unit
Costco	Membership	Item prices
Amusement parks	Entry ticket	Ride tickets
Cell phone plans	Monthly fee	Per-GB overage
Razor & blades	Razor handle	Blade refills
Printers	Printer unit	Ink cartridges
Country clubs	Dues	Green fees

Applied Insight

Welfare Effects of Price Discrimination

Does 3rd-Degree PD Raise or Lower Welfare?

Ambiguous in general! Two effects:

Misallocation Effect (–)

Output distributed inefficiently across markets. Units go to lower-value uses in the elastic market.

Output Effect (+/–)

If PD **increases total output**, welfare may rise. If total output doesn't increase, welfare falls.

Schmalensee (1981) / Varian (1985)

With linear demands, 3rd-degree PD does *not* change total output when all markets are served under uniform

Summary of Welfare Bounds

$$\Delta W \leq (P_m - MC)(Q_d - Q_m)$$

Necessary Condition for Welfare Gain

Total output must increase:

$$Q_d > Q_m$$

Applied Perspective

- **Pharmaceutical pricing:** PD across countries can improve access in low-income markets
- **Student/senior discounts:** likely welfare-improving (opens demand from price-sensitive segments)
- **Airline pricing:** complex screening; ambiguous welfare effects

Application: Price Discrimination in Pharmaceuticals

The Setting

Patented drugs are textbook monopolies: legal barriers (patents), no close substitutes, highly inelastic demand for life-saving medications. Pharma firms engage in **pervasive price discrimination**.

3rd-Degree PD Across Countries

- Identical drugs priced vastly differently: cancer biologics ~\$100K/yr in the US, ~\$30K in Europe, ~\$5K in India
- Higher prices where demand is less elastic (US — less insurance negotiation, higher income)
- No-arbitrage enforced via import

2nd-Degree PD: Airlines as Screening Masters

Mechanism	Screens For
Advance purchase discounts	Price-sensitive leisure vs. inelastic business
Saturday night stay rules	Business vs. leisure trips
Fare classes	WTP for quality
Baggage/seat fees	Unbundling — price-sensitive buy base only
Dynamic pricing	Timing reveals demand intensity

Welfare Implication

PART VI

Regulation of Monopoly

Policy responses to market power.

Regulatory Approaches

Marginal-Cost Pricing

Set $P = MC$. Allocative efficiency but may cause losses for a natural monopoly where $AC > MC$.

If AC declining at Q_{eff} : $\pi = (MC - AC)Q < 0$

Requires a government subsidy $\rightarrow$ distortions elsewhere.

Average-Cost Pricing

Set $P = AC$. Firm breaks even, no subsidy. But some DWL since $P > MC$. Classic “regulated utility” model.

Rate-of-Return Regulation

Allow firm to earn a “fair” return on

Modern Approaches

Price-Cap Regulation (RPI – X)

Prices rise with inflation (RPI) minus expected productivity gain (X).

Incentivizes cost reduction — firm keeps savings within the regulatory period.

Franchise Bidding (Demsetz, 1968)

Competition *for* the market rather than *in* the market. Auction the right to be the monopolist.

- Cable TV franchises, airport concessions, toll roads

Antitrust / Competition Policy

Natural Monopoly Regulation — Interactive

Demand (a)



Slope (b)

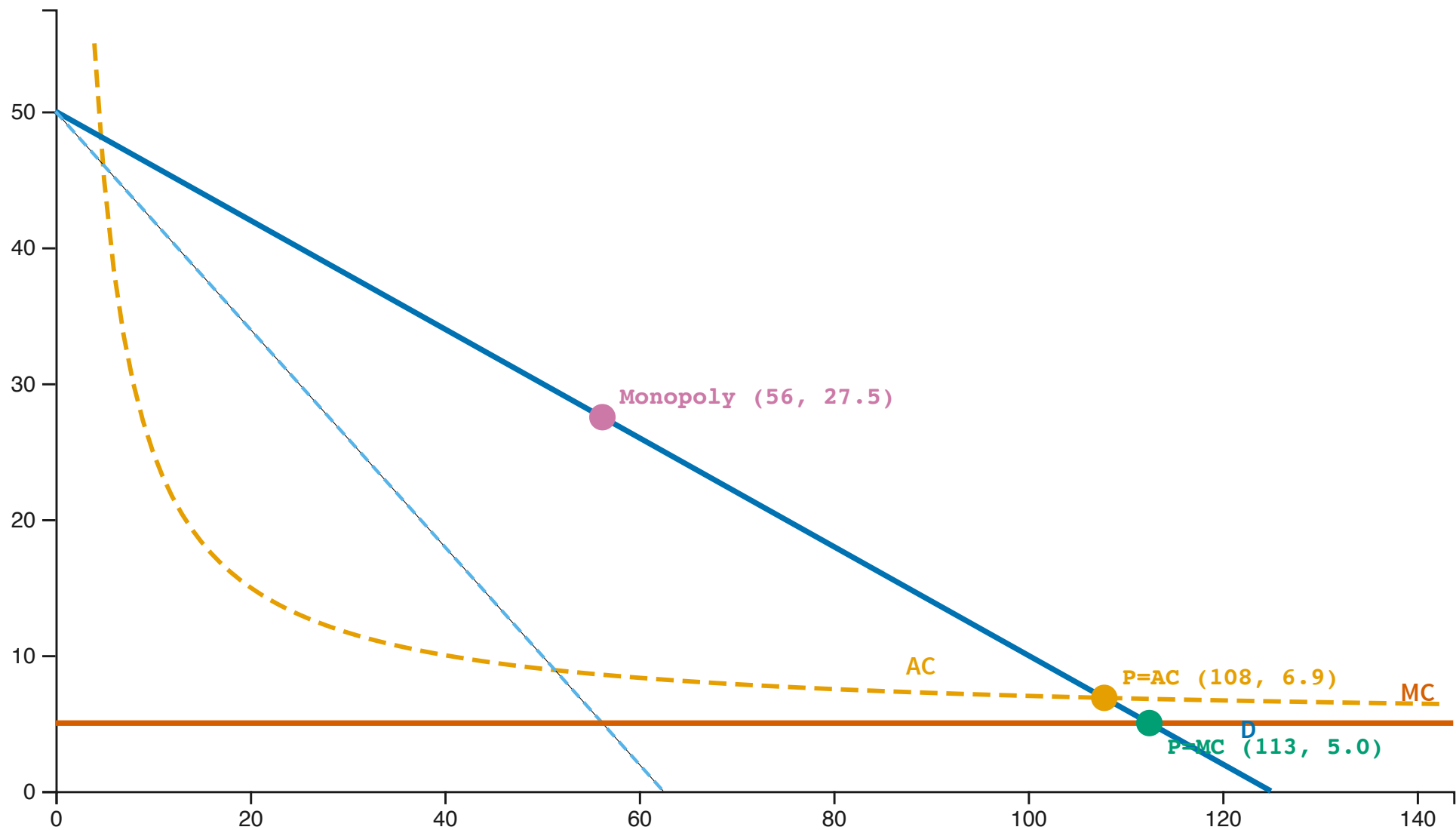


MC



Fixed Cost (F)





Monopoly: $\pi=1065.6$ | P=MC: loss=-200.0 (needs subsidy) | P=AC: $\pi=0$ (break-even)

Dynamic Views of Monopoly

Schumpeter's “Creative Destruction”

Core Argument

Monopoly profits may be the *reward* for innovation. The prospect of temporary market power drives R&D. Competition occurs *across time* through innovation, not just through price.

Suggests static DWL analysis may overstate true social cost — dynamic gains from innovation could outweigh static losses.

Arrow (1962) Counterpoint

A monopolist has *less* incentive to innovate than a competitive firm — it's

Contestable Markets (Baumol et al., 1982)

If entry and exit are **costless** (no sunk costs), even a monopolist must price at $P = AC$ to prevent “hit-and-run” entry.

Limitations

Truly costless entry/exit is rare. Sunk costs are pervasive.

Modern Synthesis

The relationship between market structure and innovation is an **inverted U** (Aghion et al., 2005):

- Too little competition → no pressure to innovate
- Too much competition → no rents to

Application: COVID-19 Vaccines and the Innovation Trade-Off

The Schumpeterian Case: It Worked

Pfizer-BioNTech and Moderna developed mRNA vaccines in **under 12 months**. Combined R&D and scale-up costs exceeded **\$10 billion**.

- Patent protection + guaranteed purchase agreements (Operation Warp Speed: \$18B) created monopoly rents needed for massive upfront investment
- Moderna earned **\$36B in cumulative vaccine revenue** (2021–22)
- This is Schumpeter in action: the *prospect* of temporary monopoly profits drove innovation that saved millions of lives

The Access Problem

By mid-2021, wealthy countries >50% vaccinated while many low-income countries <5%. The **static vs. dynamic efficiency** trade-off in stark relief:

Patents Needed

Without monopoly rents, no firm invests \$10B+ in risky R&D. Compulsory licensing ex post destroys incentives ex ante.

Access Needed

MC per dose $\approx$ \$2–5. Charging \$20–35 in poor countries creates massive DWL measured in lives. TRIPS waiver debate.

Optimal Policy?

Consent to the patent for the vaccine

Measuring Market Power: The Lerner Index

$$L = \frac{P - MC}{P} = \frac{1}{|\varepsilon_D|}$$

$L = 0$: perfect competition. $L \rightarrow 1$: extreme market power.

Practical Challenges

- **MC is unobservable** — must be estimated from cost data or conduct
- **Firm vs. market elasticity** — firm facing close substitutes has elastic residual demand even in concentrated market
- **Multi-product firms** — cross-elasticities matter

Alternative: HHI

$$HHI = \sum s_i^2$$

Empirical Estimates of Markups

Industry	Approx. Lerner
Retail food	0.05 – 0.10
Automobiles	0.10 – 0.25
Pharmaceuticals	0.40 – 0.70
Software/Tech	0.50 – 0.80
Luxury goods	0.60 – 0.85

Trend: Rising Markups

De Loecker, Eeckhout & Unger (2020): aggregate markups rose from ~1.21 in 1980 to ~1.61 in 2016. Concentrated among the largest firms. Ongoing debate: market power vs. efficiency/technology.

Summary: Key Takeaways

Core Results

- Monopolist sets $MR = MC$, charges $P > MC$
- Markup inversely related to demand elasticity: $L = 1/|\varepsilon|$
- Produces less than socially efficient quantity $\rightarrow$ DWL
- No well-defined supply curve

Price Discrimination

- 1st degree: extracts all surplus, efficient output
- 2nd degree: screening via menus, partial extraction
- 3rd degree: segment by observables; welfare depends on

Policy

- MC pricing: efficient but may need subsidies
- AC pricing: practical but some DWL
- Rate-of-return: Averch-Johnson distortions
- Price caps: better incentives but info-intensive
- Antitrust: structural vs. behavioral remedies

Looking Ahead $\rightarrow$ Lecture 7

What happens when there are *a few* firms? Oligopoly models (Bertrand, Cournot), product differentiation, strategic interaction. **Chapter 15.**